

How Crimes Affect Fear: A Data-Driven Analysis

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Abstract:

Students' perception of their safety at school is vital to a productive and healthy academic environment. The following analysis explores the relationship between how risk factors (bullying, drugs, and exposure to crime) relate to a child's fear of being attacked at school. Conclusions were drawn using the School Crime Supplement Survey conducted in 2022, interviewing adolescents across the United States about various experiences related to risks and academics. Of all of the risk factors, our studies found that verbal forms of bullying are the largest cause of fear amongst students, and are the most correlated with it. These findings suggest that schools should continue to promote anti-bullying campaigns and a positive school climate.

Introduction:

A person's childhood is often the most important and formative years of their life. Experiencing traumatic events can change a child's worldview, shaping how they view the world. If severe, these events can lead to adolescents beginning to fear for their safety. The National Center of Education of Statistics states roughly 5% of adolescents aged 12-18 fear for their safety while at school, equating to roughly 1-2 students in every classroom[1]. A child's grades are one of the most important assets to a child, as their performance often dictates their future. However, if a child is afraid of attending class for fear of an attack, their academics will suffer. Students may begin to skip classes, pay less attention, and experience stress. Therefore, the objective of this study is to identify which risks/crimes a child fears the most, so further resources can be allocated to protect adolescents from experiencing them.

Research Question:

"Which crime and school behavioral exposures are associated with the highest levels of fear among US students?"

Data & Methodology:

Dataset:

This study uses the National Center for Education Statistics' 2022 School Crime Supplement dataset[2]. The dataset samples 12335 individuals from all across the US, with 900+ pieces of information (columns) to extract from. It's one of the most extensive sources on student crime, and given that the government has also published the dataset, it is trustworthy. The survey interviews students aged 12-18 on topics such as:

- Bullying
- Fighting
- Drugs/Alcohol
- Exposure to Crime

Data Cleaning & Preprocessing:

Because the dataset was incredibly extensive and detailed, this process took time. These actions allowed the data to be tailored almost specifically to the study, ensuring only the necessary data was present. To summarize, the following steps were performed on the dataset:

- 1) Using the Codebook, select all relevant columns pertaining to the research question (900+ columns -> 41).
- 2) Handling Null Values by creating a copy of the dataset with no nulls in the target variable (student fear), and then deleting columns depending on the number of null values (if the column has >90% nulls).
- 3) Dropping duplicate entries.
- 4) Using the Codebook, Remap entries from quantitative to qualitative (e.g: survey answers of 1-4 to "Never" and "Most of the Time")
- 5) Deleting invalid entries (values such as "I don't know", or illogical answers)
- 6) Imputing mode to Null values.
- 7) Remapping entries back as integers (risk factors) to a correct, 1-4 or 0 and 1 scale (after deleting invalid entries)
- 8) Check for outliers
- 9) Revisiting deleting duplicates and imputing mode
- 10) Calculating the sum and average of each risk factor
- 11) Creating a binary classifier for Student fear. (0 = No Fear, 1 = Fear of Attack)

In the end, the final dataset had 4589 entries (students) and 45 columns.

Methods:

The analysis uses various statistical methods to uncover associations between fear and crime. Most notably, distribution analysis to observe the spread of demographics and risks/fears, correlation analysis to observe linear relationships with independent (bullying, drug use, and exposure to crime) and dependent (fear of attack at school) variables, and logistic regression to identify the most influential risk factor. Logistic regression was chosen because the question focused on association, and the target variable is binary (no/yes).

Visualizations and Analysis:

Introduction:

The following section analyzes the processed data using the statistical methods mentioned above. The analysis will be broken into three distinct categories: demographics, independent/dependent variables, and answering the research question. Each section's analysis will be paired with a handful of written insights and graphs.

Demographics:

Before beginning any analysis related to risk factors and fear, it is important to understand from whom the conclusions are drawn. This section contains the analysis of students' biological sex, age, and race.

Starting with sex, there are more males than females in the dataset. The gap between both genders is not significant (54/46), mainly because both group sizes are sufficient (2000+ entries). Therefore, while there is a slight skew in the data, it should make little to no difference in later analyses.

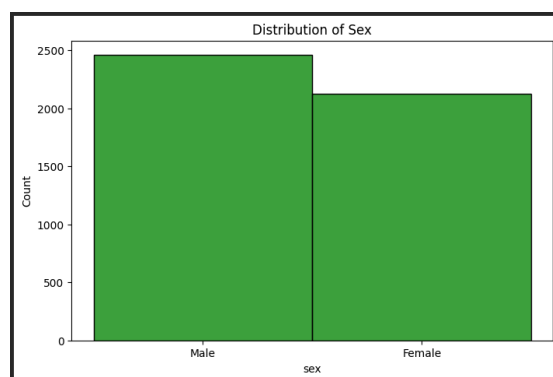


Figure 1, Barplot of Students' Sex

Additionally, given that the dataset samples the entire United States, the dataset was heavily skewed towards a White demographic. Unlike the sex distribution, the imbalance is notable, with White students accounting for roughly 70% of the student population. Given roughly 50% of the nationwide student population is White [3], the sample over-represents the White population. While class imbalance is typically an issue, logistic regression helps control confounding variables such as race and calculate risks/fear independent of a student's background.

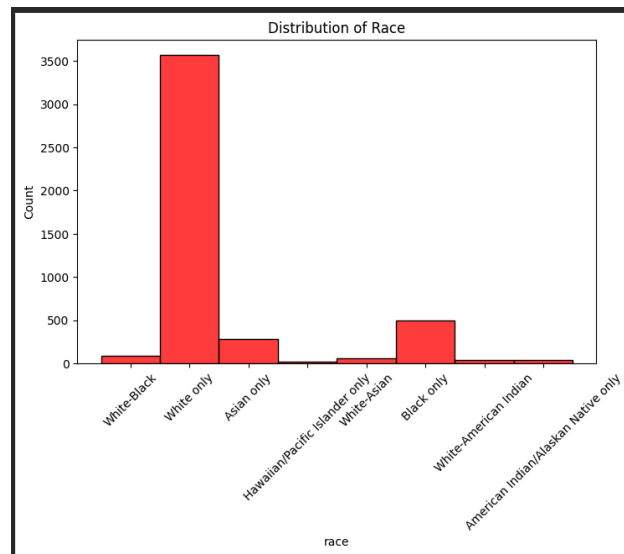


Figure 2, Distribution of Student Races.

Lastly, student ages seem to be fairly distributed. While some age groups have more counts than others, for the most part, students are uniformly distributed. However, there seems to be a 50% reduction in 18-year-olds compared to other ages. Because not all high school graduates are 18 before commencement, there are significantly fewer of their age group.

Furthermore, the age distribution shows the survey has a slight bias towards high schoolers compared to middle school students. Because most American middle schools are 3 years long, while high schools are 4, the survey's sampling structure resulted in more high schoolers being sampled.

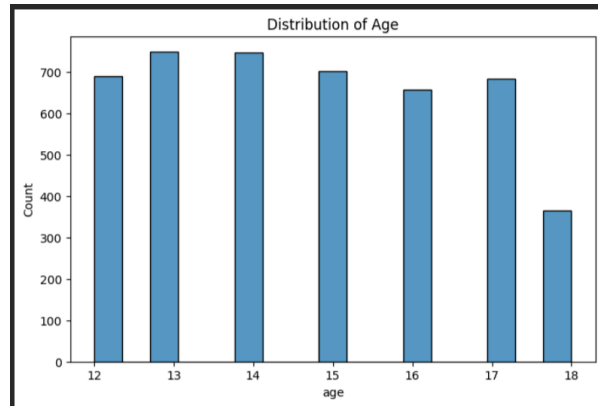


Figure 3, Distribution of Student Ages.

Risk Factors:

With a firm understanding of the dataset's audience, it's time to uncover each risk factor and explore its trends and patterns (individually, and in relation to others). Each Risk factor has a sum (total number of experiences) and a mean (overall score from 0 to 1), allowing us to calculate each risk's distribution and use its mean in subsequent statistical analyses. All factors were pulled from the dataset and assessed into one of 4 categories:

- 1) Verbal Bullying
- 2) Physical Bullying
- 3) Drug/Alcohol Availability
- 4) Exposure to Crime

Before understanding the distribution of each risk factor, it's important to see the bigger picture. The study observed the total risk factors experienced by all students. Fortunately, it seems most students have experienced at most one risky event, with the maximum threshold on the boxplot being 7 events. However, this also means the data is heavily right-skewed. As a result, there are few highly informative outliers, the primary information desired. However, Similar to how the model addresses racial demographic imbalances, logistic regression thankfully handles this issue by using the maximum likelihood estimation, adapting to the skewed probabilities.

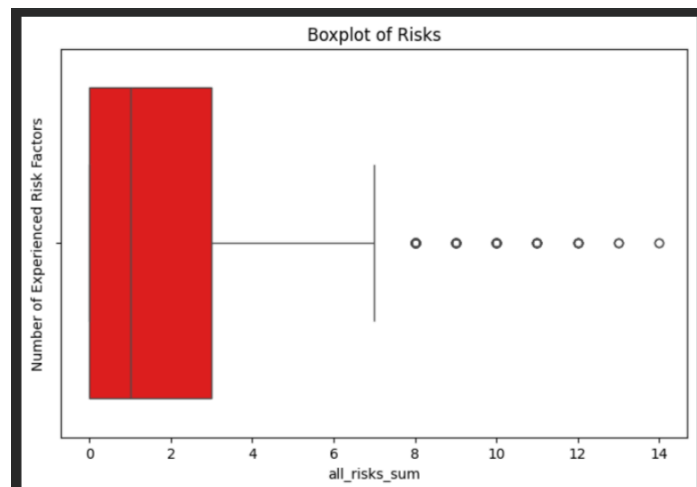
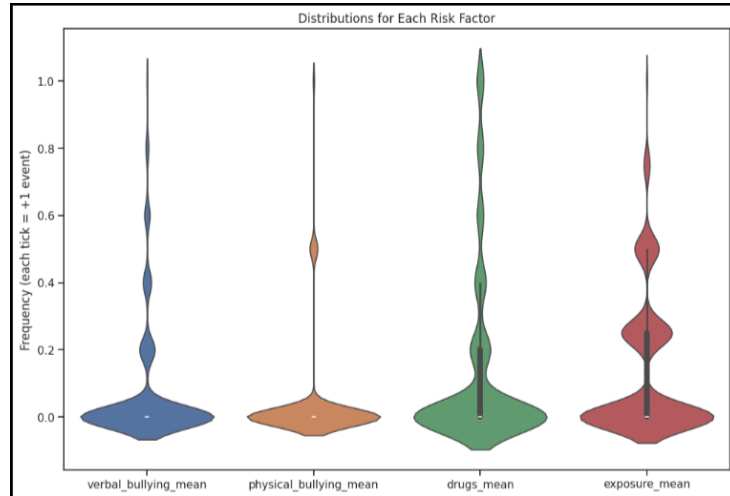


Figure 4, Distribution of Student Experiences with "Risk" Events.

After observing the skewness in the sum of risk factors, it's unsurprising that each risk factor's distribution is just as lop-sided. However, it seems that students are more exposed to seeing drugs and crime than being bullied. It's expected that more students will have been exposed to environmental exposures (drugs/ crime) than to interpersonal victimization (bullying).



Note: (0 = No experience, 1= Experienced all in category).

Figure 5, Distribution of Each Risk Factor.

Having seen the distribution of both the summation of all risk factors and the means of each, it goes without saying that the number of students who fear being attacked at school is just as one-sided as the others. roughly 95% of students do not fear an attack at school, while the other 5% do.

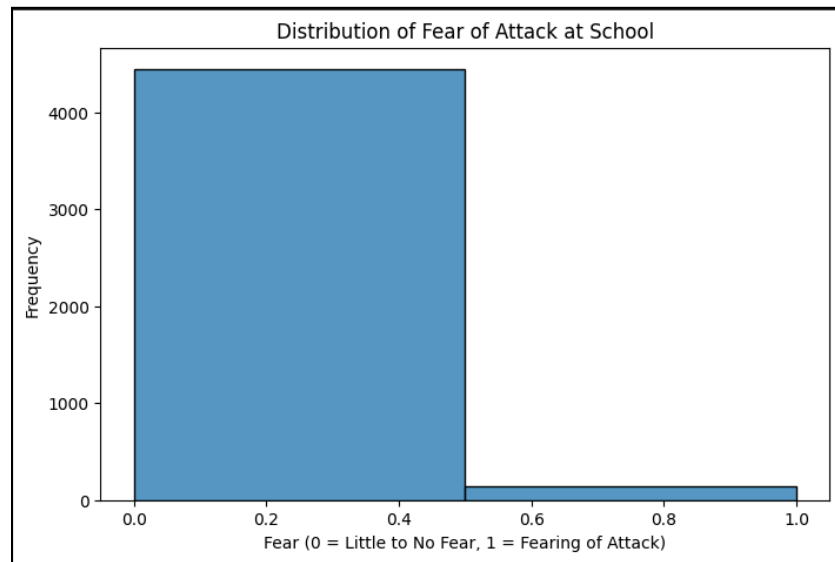


Figure 6, Distribution of Students who Fear an Attack at School.

Research Question:

The following visualizations will now begin to answer the research question. This will be done by observing the correlations between crime factors and fear, as well as calculating odds ratios using logistic regression.

Despite believing there would be a strong correlation between the risk factors and fear, the correlation graph paints a different picture. The highest correlation coefficient is 0.33, obtained through verbal bullying, with physical bullying scoring 0.28. This tells us one of two pieces of information: the relationships between these variables are non-linear, or given that the target variable is binary, the correlation coefficient shrinks as a result (often takes continuous, normally distributed data). Again, this issue will be remedied in the upcoming logistic regression analysis, thanks to its more robust calculation of probabilities.

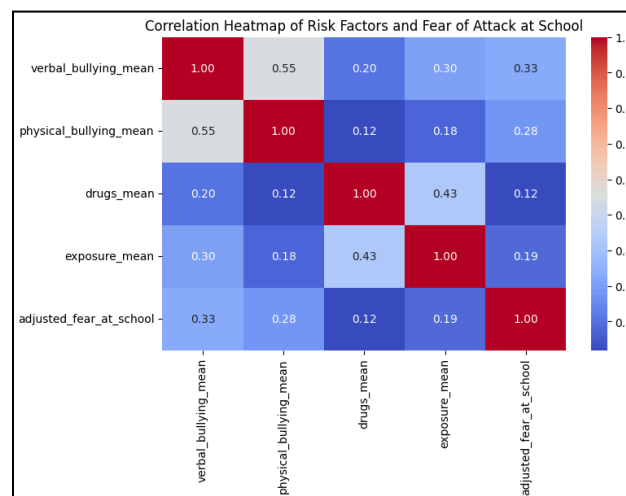


Figure 7, Heat Map of Risk Factors and Fear of Being Attacked at School.

Finally, logistic regression was implemented on the data, using the means of all risk factors and a binary classifier of whether a student fears being attacked or not. Analysis was performed on the dataset using a 70/30 training/testing split, visualizing the ROC/AUC curve and the odds ratios.

First, the ROC/AUC curve shows the model is being trained using the data, being roughly 82% correct at classifying whether a student is afraid of being attacked. As a result, we can confidently conclude that the risk factors do influence whether or not a student fears being attacked.

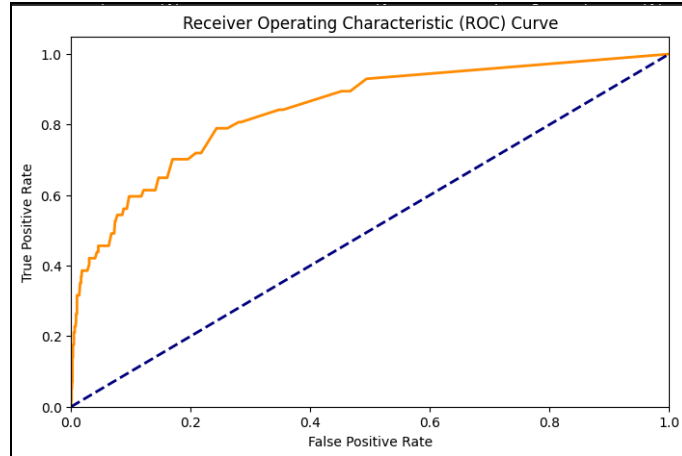


Figure 8, ROC/AUC Curve of the Fitted Logistic Regression Model.

The stark difference between verbal abuse's impact on student fear compared to the other risk factors is shocking. Unlike the correlation coefficients, verbal forms of bullying are clearly the largest factor causing student fear of attacks. With a ratio of 17, this means students experiencing verbal abuse at school have 17x higher odds of 1 an attack at school than a student who was not verbally attacked. This is likely because implying a threat and making a student feel unwelcome in an environment may cause them to fear being eventually targeted or physically attacked. Students don't know if they will be attacked, and therefore are more likely to be wary of their surroundings.

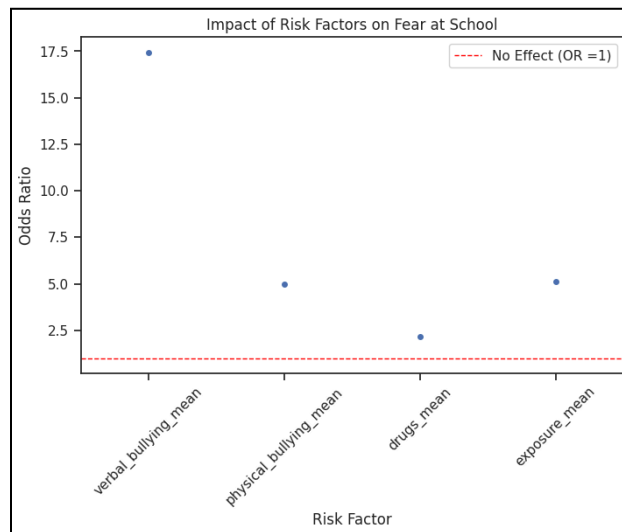


Figure 9, Odds Ratios of all Risk Factors in Association with Fear at School.

Conclusion:

The following project aimed to answer which crime and school behavioral exposures are associated with the highest levels of fear among US students, answering the question methodically using exploratory data analysis and statistical testing. Of the following factors, verbal abuse seemed to be the most effective and had the highest correlation/association with fear. Taking the necessary precautions to prevent verbal abuse from happening will likely greatly increase student morale, encouraging attendance and school safety.

Examining this dataset truly highlights the idea clearly: Bullying is the primary driver for adolescents' fears. It's not the exposure to crime or other outside influences; it's the environment students are in that causes them to fear the people within. While not all too surprising, the odds ratios for all risk factors demonstrate how hurtful and effective verbal abuse/bullying can be. Words can lead to long-lasting psychological trauma, and (in this case) cause a child to be fearful for their safety. Schools need to continue to discourage bullying and protect students who are suffering from it, as doing so can foster a healthy environment free of fear and safety concerns.

Resources:

[1] "COE - Students' Perceptions of Personal Safety at School and Away from School." Nces.ed.gov, nces.ed.gov/programs/coe/indicator/a16.

[2] "School Crime Supplement (SCS)." Bureau of Justice Statistics, 2022, bjs.ojp.gov/data-collection/school-crime-supplement-scs#1-0.

[3] National Center for Education Statistics. "Racial/Ethnic Enrollment in Public Schools." Nces.ed.gov, May 2024, nces.ed.gov/programs/coe/indicator/cge/racial-ethnic-enrollment.